

COMPUTER NETWORKS

Assignment

Static Routing, Network Devices
and Random Access Protocols

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Q1. Static Routing — 4 Routers, 4 Switches & 8 PCs

[6 Marks]

The network has four routers (Router4, Router5, Router6, Router7) connected to each other in a chain using serial WAN links. Each router is also connected to a switch, and each switch connects two PCs, making four separate LAN networks in total. Static routes are configured on each router so that every PC can communicate with every other PC across the network, even though they are on different networks.

LAN Networks and Network IDs:

- Network 192.168.1.0 — Router4, Switch0, PC0, PC1
- Network 192.168.2.0 — Router5, Switch1, PC2, PC3
- Network 192.168.3.0 — Router6, Switch2, PC4, PC5
- Network 192.168.4.0 — Router7, Switch3, PC6, PC7

WAN (Router-to-Router) Networks:

- Network 192.168.5.0 — between Router4 and Router5
- Network 192.168.6.0 — between Router5 and Router6
- Network 192.168.7.0 — between Router6 and Router7

IP Addresses (Text Format):

- Router4 — Gi0/0: 192.168.1.3, Se0/0/0: 192.168.5.1
- Router5 — Gi0/0: 192.168.2.3, Se0/0/0: 192.168.5.2, Se0/0/1: 192.168.6.1
- Router6 — Gi0/0: 192.168.3.3, Se0/0/0: 192.168.6.2, Se0/0/1: 192.168.7.1
- Router7 — Gi0/0: 192.168.4.3, Se0/0/0: 192.168.7.2
- PC0: 192.168.1.1 PC1: 192.168.1.2
- PC2: 192.168.2.1 PC3: 192.168.2.2
- PC4: 192.168.3.1 PC5: 192.168.3.2
- PC6: 192.168.4.1 PC7: 192.168.4.2

Sample Static Route (on Router4, to reach the other networks):

- ip route 192.168.2.0 255.255.255.0 192.168.5.2
- ip route 192.168.3.0 255.255.255.0 192.168.5.2
- ip route 192.168.4.0 255.255.255.0 192.168.5.2

Similarly, static routes are added on Router5, Router6 and Router7 pointing to the correct next-hop address so that all four networks can reach one another.

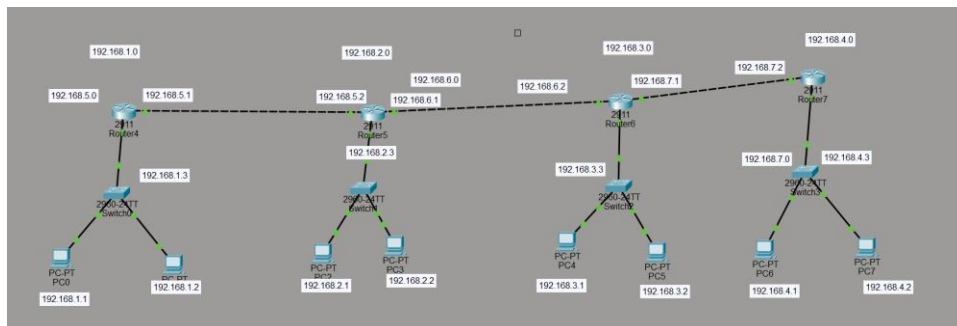


Fig 1.1 — Cisco Packet Tracer topology for static routing

Q2. OSI Model

[3 Marks]

The OSI (Open Systems Interconnection) Model is a 7-layer reference model that explains how data travels from one computer to another over a network. Each layer has its own job and passes data to the next layer. Below are the seven layers along with their functionality, data format and common protocols.

1. Physical Layer

Functionality: Sends raw bits (0s and 1s) over the physical medium such as cables or radio signals; deals with voltages, connectors and cabling.

Data Format: Bits

Protocols: Ethernet (physical), USB, Bluetooth

2. Data Link Layer

Functionality: Provides node-to-node delivery on the same network, physical (MAC) addressing, framing, and basic error detection.

Data Format: Frame

Protocols: Ethernet, PPP, HDLC

3. Network Layer

Functionality: Handles logical (IP) addressing and routing — deciding the best path for data to reach a different network.

Data Format: Packet

Protocols: IP, ICMP, ARP, OSPF, RIP

4. Transport Layer

Functionality: Provides end-to-end delivery of data between hosts, including segmentation, flow control and error control.

Data Format: Segment

Protocols: TCP, UDP

5. Session Layer

Functionality: Opens, manages and closes a communication session (dialogue) between two devices.

Data Format: Data

Protocols: NetBIOS, RPC, PPTP

6. Presentation Layer

Functionality: Converts data into a common format the application can understand; handles encryption and compression.

Data Format: Data

Protocols: SSL/TLS, JPEG, MPEG

7. Application Layer

Functionality: The layer closest to the end user; provides network services directly to applications like browsers and email clients.

Data Format: Data

Protocols: HTTP, HTTPS, FTP, SMTP, DNS

Q3. Error Detection Methods

[3 Marks]

While transmitting data over a network, noise or interference in the channel can flip bits and corrupt the message. Error detection techniques allow the receiver to check whether the received data has been altered during transmission. The three common methods are Parity Check, Checksum and Cyclic Redundancy Check (CRC).

1. Parity Check (Vertical Redundancy Check)

An extra bit, called the parity bit, is added to the data so that the total number of 1s becomes even (even parity) or odd (odd parity). The receiver counts the number of 1s and compares it with the expected parity; a mismatch indicates an error.

Example: Consider the 7-bit data 1011001 (four 1s) using even parity. Since the count of 1s is already even, the parity bit added is 0, giving the transmitted byte 10110010. If, during transmission, one bit is flipped and it is received as 10110000 (three 1s), the receiver recounts the 1s, finds the total is odd, and detects that an error has occurred.

2. Checksum

The data is divided into fixed-size blocks. At the sender, all blocks are added together using 1's complement arithmetic, and the complement of this sum is sent along with the data as the checksum. At the receiver, the same addition (including the checksum) is performed; if the result is all 1s, the data is accepted as error-free, otherwise it is rejected.

Example: Two 8-bit blocks 10101001 and 00111001 are sent. The sender adds them to get 11100010, and sends its complement, 00011101, as the checksum. The receiver adds the two data blocks and the checksum together; if the total comes out as all 1s, the data is correct, otherwise an error is flagged.

3. Cyclic Redundancy Check (CRC)

CRC divides the data bits by a fixed, pre-agreed generator polynomial using binary (XOR) division. The remainder of this division is appended to the data as the CRC before transmission. At the receiver, the same division is performed on the received frame; a remainder of zero means the data is error-free, while a non-zero remainder means an error occurred.

Example: Let the data be 1101011011 and the generator be 10011. Four zeros are appended to the data before dividing, and XOR division gives a remainder, say 1110, which becomes the CRC. The frame 11010110111110 (data + CRC) is transmitted. The receiver divides the entire received frame by 10011; a zero remainder confirms the frame arrived without error.

Comparison:

- Parity Check — simplest method, but can miss errors when an even number of bits are flipped.
- Checksum — used at higher layers (e.g., TCP/IP) and can detect most burst and random errors.
- CRC — most reliable and widely used at the Data Link layer (e.g., Ethernet); detects almost all transmission errors.